Worksheet 5.2

Balancing chemical equations

Student: .... Class: ....

1. Word equations and formula equations can be used to describe a chemical reaction.

Example

Word equation: methane + oxygen → carbon dioxide + water

Formula equation: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Check that the number of atoms of each element is the same for the reactants and products.

Use the tabulated information to construct formula equations from the supplied word equations.

Name	Formula	Name	Formula	Name	Formula
Calcium oxide	CaO	Hydrochloric acid	HCl	Ethane	C_2H_6
Sodium carbonate	Na_2CO_3	Nitric acid	HNO_3	Acetylene	C_2H_2
Calcium nitrate	$Ca(NO_3)_2$	Sodium chloride	$NaCl$	Calcium carbonate	$CaCO_3$

	Formula equation
Calcium oxide + nitric acid → calcium nitrate + water	
Sodium carbonate + hydrochloric acid → sodium chloride + carbon dioxide + water	
Ethane + oxygen gas → carbon dioxide + water	
Acetylene + oxygen → carbon dioxide + water	
Calcium carbonate → calcium oxide + carbon dioxide	

2. Iron (III) nitrate ($Fe(NO_3)_3$) solution reacts with sodium hydroxide ($NaOH$) solution to form a brown precipitate of iron (III) hydroxide ($Fe(OH)_3$). Sodium nitrate ($NaNO_3$) remains in solution as it is a soluble compound.

Write the word equation and the balanced formula equation for this reaction.

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